



The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	DDA6001
Course title (English)	Advanced Stochastic Process
Course title (Chinese)	高级随机过程
Units	3
Description (English)	A stochastic process is a mathematical model for random phenomena which change in time. With examples such as coin tossing, betting, queueing system, branching process, and stochastic networks, this course introduces a basic theory for modelling stochastic processes. It intends to help students to perform applied and theoretical studies on stochastic processes. Topics include martingales, Poisson processes, Markov processes, Brownian motions and stochastic differential equations. A basic probability course is prerequisite for this course.
Description (Chinese)	介绍随机过程基本理论和各种应用。主题包括马尔可夫过程，更新理论，随机游动，分支过程，布朗运动，平稳过程，鞅和点过程，以及排队理论和库存模型的应用。

B. Corresponding course at CUHK

Please give details of the *closest* corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or



any other requirements.

A. Prerequisites

B. Co-requisites

C. Others

3. Learning outcomes

1. Be able to design a mathematical model for random phenomena
2. Brush up computation skills for basic probability analysis
3. Be able to use basic tools for stochastic analysis such as renewal theorem, stochastic basis, conditional expectation, martingale and optional sampling theorem
4. Understand the framework for discrete-time stochastic processes
5. Understand the measurability framework for continuous-time processes
6. Widen knowledge about Poisson processes, Brownian motions, stochastic differential equations and Markov processes.
7. Be able to apply stochastic analyses for various applications such as queueing, inventory, communication networks, risk process and system management
8. Be able to do theoretical studies for stochastic processes and their applications

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

Introduction to a probability model

Limit of a sequence and its summation, coin tossing, discrete general probability spaces, random variable, expectation, test function, moment, law of large numbers, central limit theorem, mode of convergence for a sequence of random variables

Discrete-time Markov chain and renewal process

State classification, limiting behaviors, stationary distribution, random walk, queueing model, branching process, counting process, renewal theorems

Density based approach for continuous-time process

Joint density, conditional density, Poisson process, Brownian motion

Mathematical framework for discrete-time process



Construction of a probability space, metric space, open set, separability, Borel field, filtration, conditional expectation, martingale, Doob decomposition, semi-martingale, stopping time, optional sampling, Markov process

Martingale limit theorems in discrete-time

Convex function, Jensen's inequality, super- and sub- martingales, up-and-down crossing, martingale convergence, uniform integrability, and their applications

Mathematical framework for continuous-time processes

Measurable space for continuous-time processes, optional and predictable σ -fields and their processes, stopping time, stochastic interval, local martingale, optional sampling

Examples of continuous-time processes and martingale characterizations

Poisson process, continuous-time Markov chain, Brownian motion

Overview on stochastic analysis

Time evolution of a process, semi-martingale, Doob-Meyer decomposition, stochastic integral by martingale, quadratic variations, Ito integration formula

5. Assessment scheme

Component/ method	% weight
Assignment	20
Midterm	30
Final	50

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type: **General**

Grade	Description
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner. Specifically, outstanding performance on learning outcomes 1-10.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and



	comprehensive. Has the ability to express the knowledge or application with clarity. Specifically, generally outstanding performance on learning outcomes 1-6, substantial performance on learning outcomes 7-10.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way. Specifically, substantial performance on learning outcomes 1-6, acceptable performance on learning outcomes 7-10.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way. Specifically, satisfactory performance on learning outcomes 1-6, acceptable performance on learning outcomes 7-10.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials. Has the ability to state the knowledge or application in simple terms. Specifically, barely satisfactory performance on learning outcomes 1-6.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements. Specifically, failure to achieve barely satisfactory performance on learning outcomes 1-5

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

NA

8. Readings

9. Course components

Activity	Hours/week
Lecture	3



Tutorials	1
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10. Indicative teaching plan

Week	Content/ topic/ activity
1	Review of basic probability theory
2	Review of basic probability theory
3	Brownian motions
4	Brownian motions
5	Stochastic integrals
6	Stochastic integrals
7	Stochastic differential equations
8	Stochastic differential equations
9	Stochastic differential equations
10	Applications
11	Applications
12	Poisson process, queueing theory, continuous-time Markov chain
13	Poisson process, queueing theory, continuous-time Markov chain
14	Poisson process, queueing theory, continuous-time Markov chain

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval



Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

14. Any other information

15. Version date

Version number	1
As of (date)	2025/11/13